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Sample program for Nested if else based on college grade system.

For -> Entry check loop

For, else

For, while

Write the nested loop

1) for while, while for.

Nested loops - takes time

Break
Pause
Continue
} Jumping statement or loop control statement.

Break.

(1) Write a program that asks the user to input numbers until they input 0. The program should print the sum of all the input numbers.

Continue.

Write a program that prints all the numbers from 1 to 10 except 5 using a for loop and continue statement.

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datasets, batch data

List [ ] - Ordered, mutable, allows duplicates, Use for

Tuple ( ) - " " , Use for model dimensions (28, 28, for image shapes

Set { } - Unordered, no duplicated. Use to find unique

Dict { : } -

Functions - Reusable code

Functions is a grp of statement that performs a specific task.

Types

* User defined - Defined by the user themselves
* Build in - In built within py.
* Recursion - That calls itself.
* Lamda - Unnamed function

Task : Write a function to calculate CGPA

Date

-> Collect the data

-> Data cleaning.

structured data

Order, proper format

Unstructured data

Mixing of Random data

Correcting errors

-> Data transformation

Preparing the data for analysis

-> Data exploration (Pandas, Numpy)

exploring data analysis

helps to identify the trends, patterns

-> Data visualisation

-> Data interpretation.

EDA
Summarising, visualising and understanding the structure of data
The libraries are
pandas
numpy
matplotlib
scikit learn
seaborn

Pandas:
Data manupulation and analysis
It have 2 catagories
1D - series, 2D - data frame

pip is a package manager
command to install pandas is
pip install pandas.

Numpy:
It is a numerical python, It's for manipulating large data types.
Numpy arrays are multidimensional

Matplotlib and seaborn
Primary libraries for visualisation

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Matplotlib is a highly visualisation data.

Seaborn is high level interface for creating statistical interface. It simplifies the process of complex visualisation

Scikit learn is a machine learning library that provides tools for classification, clustering, statitics

OS Module

to manage files

r, w, x - create, a - append, b, t, +

read & write

r - Error if file doesn't exist.
w - Overwrites if file exists
x - Error if file exists
a - writes at the end
b - for images, executables
t - handles encoding / decoding.
'r+', 'w+'

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Creating a Directory.

To create a directory, use the
os.mkdir('new-directory')

To create nested directories using
os.makedirs():
os.makedirs('parent-directory/child-directory')

Text file:
Reading files:

```
with open('new-file.txt','r') as file:
content = file.read()
print(content)
```

To overwrite a file, use the 'w' mode:

```
with open('new-file.txt','w') as file:
file.write('New content!')
```

Renaming files

```
print("\n Rename 'doc1.txt' to
'document_001.txt'")
os.rename('doc1.txt', 'document_001.t
print("Renamed file:")
print(os.listdir('.'))
```

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```
import os
import shutil
import glob

# current coorking directory
currend_dir = os.getcwd()
O/P = c:\users\~~~

# list files and folders in current directory
print("\nlist files and folders")
items = os.listdir('.')
print(items)

# Create file
print("\n Create sample files")
test_files = ['doc1.txt', 'image1.jpg', 'data.
                                        csv']
for file in test_files:
    with open(file, 'w') as f:
        f.write("Sample content for "+file)
print(f"Created files : {test_files}").
```

ML

-> Supervisor ML & Unsupervisor ML

Supervisor ML
↓
Target
↓
Already classified
↓
The algorithms are recursion and clustering

Unsupervisor ML
↓
No target
↓
We have to classify the data
KNN, SVM, Classification.

Variables or data
Numerical variables (Quantitative)
It represent numerical values
Categorical variable (Qualitative)
Eg: Region, Country

Supervisor ML
Eg: Filtering spam or nonspam emails
Unsupervisor ML
eg: To classify the ticket prices based upon department.

Define -> Collect -> prepare -> Build
Evaluate -> Deploy & use

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Recursion -> to find the relnship btwn input & output variable, or dependt & indepn

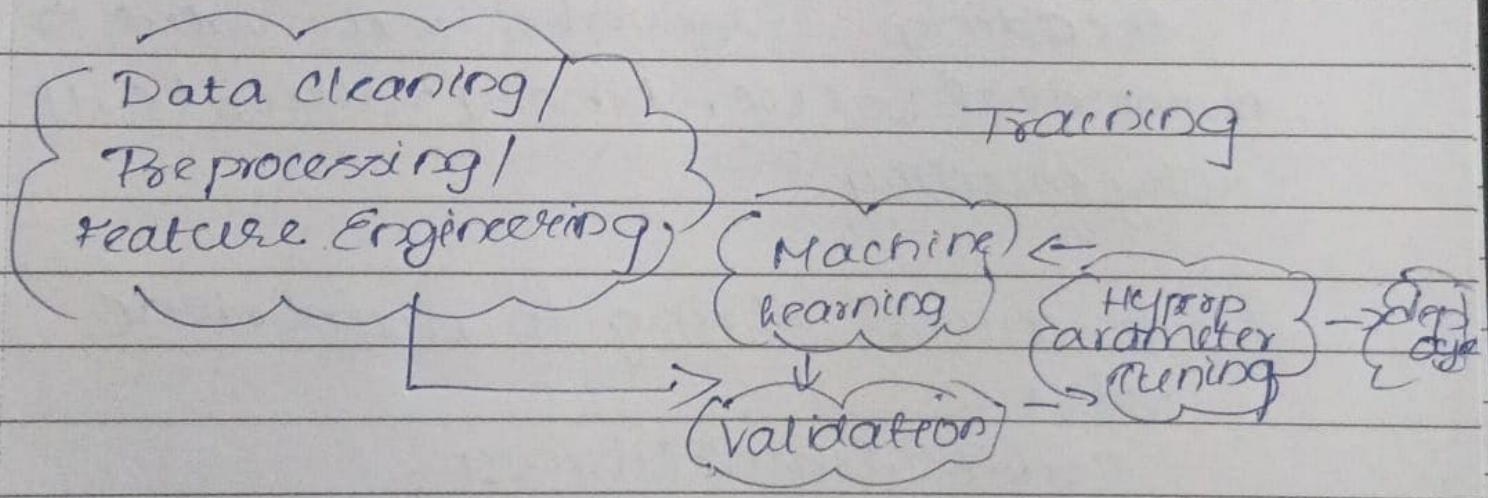
Two types, Linear & Logistic

Linear -> Straight line (Decimal values)
We have to predict, Eg: Rainfall

$y = mx + c$

Logistic -> Eg: Loan, Boolean Algebra.

ML FLOW




```
import pandas as pd
df = pd.read_csv('    ')

df

Po calculating
```

Handling missing values

```
mu = df ['Income'].mean()
```

To identify the

```
mu = df ['Income'].mean()
mu
```

To identify catagories

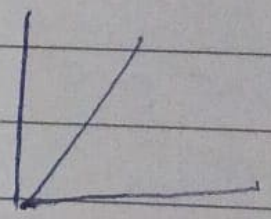
```
df ['Employment_Type'].value_counts()
```

Encoding -> converts text value to numerical value, binary values (0,1)

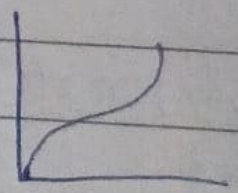
Why encoding?

To convert string to numerical value.

Scikit learn library



Linear recursion



Sigmoid recursion

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hyperparameter tuning

before

1	0.89	0.79	0.82	170

After

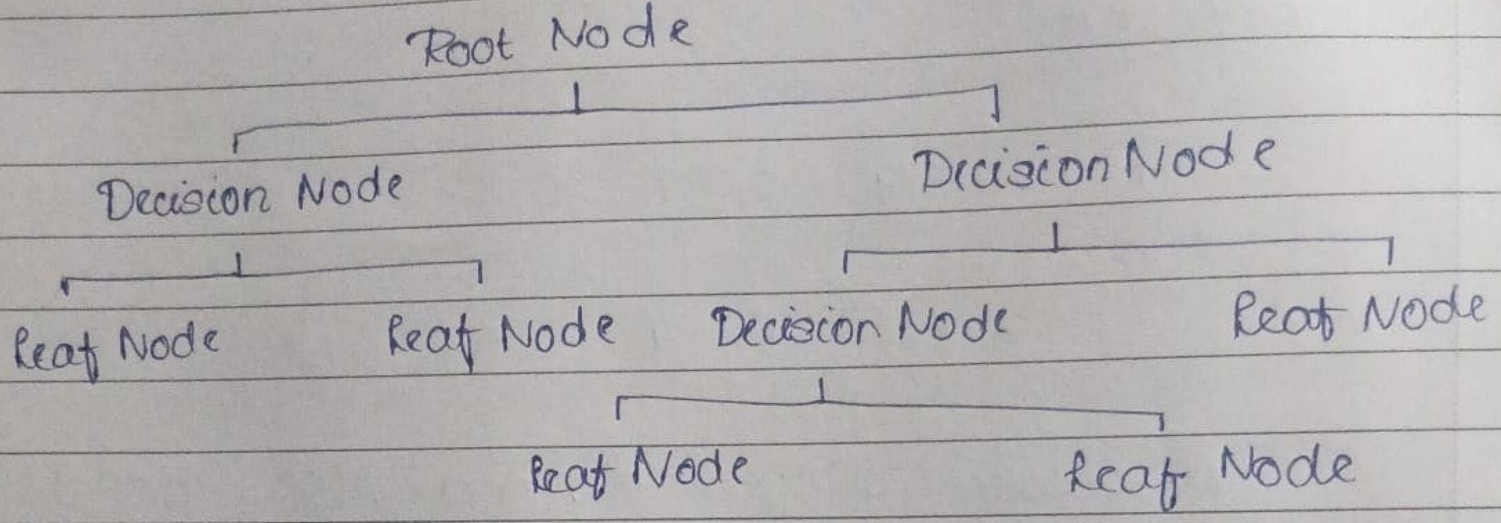
	0.56	0.59	0.56	200

```
lr = LogisticRegression(max_iter=1000)

import matplotlib.pyplot as plt
plt.bar(df["Age"], df["Income"])

plt.xlabel("Age")
```

RFC → combination of 1 or more decision tree / balanced algorithm

Used in predictive AI

Decision tree




```
from sklearn.ensemble import RandomForestClassifier

rf_model =
```

{Task 3}

Write a ML model to predict the student's next sem marks, based upon last sem and test marks.

Date

1. choose the no. of clusters k
2. Initialize k centroids (randomly select points)
3. Assign each data point to the nearest centroid
4. Recalculate the centroids (take mean of each cluster)
5. Repeat steps 3 and 4 until the clusters do not change or centroids stabilize.
6.

The basic analytical tools are.

Excel

Tableu

power BI

python

R programing

SQL

Unsupervised learning

Discover hidden patterns and structures in data tout the need

clustering (k-means)

To grp similar data

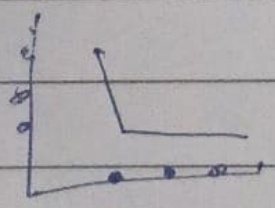
Stps

choose the grps.

Initialize (rand

After clustering

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Elbow technique

K = 4